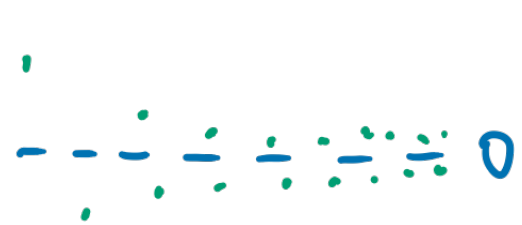


Math 421

Monday, October 27

Announcements

- Grace's Drop-In Hours
 - Wednesday 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Homework 7 Due Friday



$$s_n = (-1)^n \frac{1}{n} \quad \liminf s_n = \limsup s_n$$

$\{n : \liminf s_n = s_n = \limsup s_n\}$ is empty.

Activity

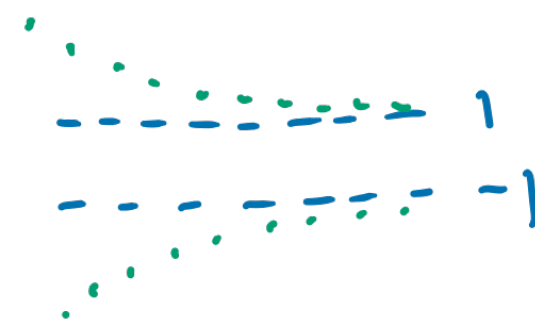
Give sequences that exhibit the following properties:

❖ The set $\{n : s_n > \limsup s_n\}$ can be infinite. $s_n = \frac{1}{n} \quad \limsup s_n = 0$

❖ The set $\{n : s_n < \liminf s_n\}$ can be infinite. $s_n = -\frac{1}{n} \quad \liminf s_n = 0$

❖ If $\liminf s_n < \limsup s_n$, the set

$$\{n : \liminf s_n \leq s_n \leq \limsup s_n\}$$



can be empty. **Hint:** Try to sketch out what this property is saying, what would need to be true about a sequence that satisfies this property?

$$s_n = (-1)^n \left(1 + \frac{1}{n}\right)$$

$\liminf s_n \in S$
 $\limsup s_n \in S$

$\liminf$, $\limsup$ and the set of subsequential limits $= S$

Theorem 11.7. Let (s_n) be any sequence. There exists a subsequence whose limit is $\limsup s_n$, and there exists a subsequence whose limit is $\liminf s_n$. - Proof uses Thm 11.2.

Theorem 11.8. Let (s_n) be any sequence in $\mathbb{R}$, and let S denote the set of subsequential limits of (s_n) .

1. S is nonempty.
2. $\sup S = \limsup s_n$ and $\inf S = \liminf s_n$. $\star$
3. $\lim s_n$ exists if and only if S has exactly one element, namely $\lim s_n$. $\star$

$\liminf$

$\limsup$

Examples

1. (s_n) where $s_n = \frac{1}{n}$. $S = \{0\}$
2. (a_n) where $a_n = n^2(-1)^n$. $A = \{-\infty, +\infty\}$.
3. (b_n) where $b_n = \sin\left(\frac{n\pi}{3}\right)$. $B = \left\{-\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2}\right\}$.
4. (r_n) a list of all rational numbers. $R = \mathbb{R} \cup \{-\infty, +\infty\}$.
5. (c_n) where $c_n = n(1 + (-1)^n)$. $C = \{0, +\infty\}$.

Chapter 2: Sequences

Section 10 (Part 3): Cauchy Sequences

$$(s_n) \quad s_n = \frac{1}{n} \quad \epsilon = \frac{1}{2} \quad \text{there exists } N = 10$$

such that $n, m > N$

$$\left| \frac{1}{n} - \frac{1}{m} \right| < \frac{1}{2}$$

$$\left| \frac{1}{11} - \frac{1}{12} \right| < \frac{1}{2}, \quad \left| \frac{1}{13} - \frac{1}{50} \right| < \frac{1}{2}$$

Cauchy Sequences

Def: A sequence (s_n) of real numbers is called a *Cauchy sequence* if for each $\epsilon > 0$ there exists a number N such that

for all $n, m > N$ we have $|s_n - s_m| < \epsilon$.

Q: What is this definition saying in your own words? How does this definition compare to the definition of a convergent sequence?

- beyond a certain N value the terms of the sequence become arbitrarily close to each other.
- s_n is not a constant, and not the limit.

Convergent $\Rightarrow$ Cauchy

Lemma 10.9. Convergent sequences are Cauchy sequences.

Activity: Sketch a proof of the Lemma.

① What are you given?

for all $\varepsilon > 0$ there exists N such that for all $n > N$
 $|s_n - s| < \varepsilon/2$ $\leftarrow$ allowed to modify

② What are you trying to show?

for all $\varepsilon > 0$ there exists N such that for all $m, n > N$
 $|s_n - s_m| < \varepsilon$ $\leftarrow$ remains arbitrary.

③ Sketch the details. let $\varepsilon > 0$. Since (s_n) is convergent, $\lim s_n = s$, there exists N such that for all $n > N$, $|s_n - s| < \varepsilon/2$. Assume $n, m > N$, then.

$$|s_n - s_m| = |s_n - s + s - s_m| \leq |s_n - s| + |s_m - s| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Cauchy $\Rightarrow$ Bounded

Lemma 10.10. Cauchy sequences are bounded.

Proof. Let (s_n) be a Cauchy sequence. Then there exist $N \in \mathbb{N}$ such that for all $n, m > N$, $|s_n - s_m| < 1$.
So $|s_n - s_{N+1}| < 1$ then for all $n > N$ $|s_n| < |s_{N+1}| + 1$.
Set $M = \max \{|s_1|, |s_2|, \dots, |s_N|, |s_{N+1}| + 1\}$ then
 $|s_n| \leq M$ for all $n \in \mathbb{N}$ and so (s_n) is bounded.